

SQL RETURNING Clause

Important: RETURNING is not a core MySQL feature

It is a first-class feature in PostgreSQL, SQLite (recent versions), Oracle, and SQL Server (through OUTPUT)

Why RETURNING Exists

Normally, when you execute an INSERT, UPDATE, or DELETE, SQL tells you only how many rows were affected.

Example:

```
UPDATE Employees
SET Salary = Salary * 1.10
WHERE EmployeeID = 5;
```

Output:

```
Query OK
1 row affected
```

But what if you immediately want to know:

- Which row was inserted?
- What is the generated ID?
- What is the updated salary?
- Which row got deleted?

Without RETURNING, you often need an additional SELECT.
RETURNING combines both operations.

PostgreSQL

```
INSERT INTO Students(Name, Age)
VALUES ('Aadya', 20)
RETURNING *;
```

Output

StudentID	Name	Age	CreatedAt
101	Aadya	20	2026-07-18 10:15:20

Returning only specific columns

```
INSERT INTO Students(Name, Age)
VALUES ('Aadya', 20)
RETURNING StudentID;
```

Output

StudentID
101

Return selected columns

```
UPDATE Employees
SET Salary = Salary + 5000
WHERE EmployeeID = 5
RETURNING EmployeeID, Salary;
```

MySQL

Normally you do:

```
UPDATE Employees  
SET Salary = Salary + 5000  
WHERE EmployeeID = 5;
```

Then

```
SELECT *  
FROM Employees  
WHERE EmployeeID = 5;
```

PostgreSQL

```
DELETE FROM Employees  
WHERE EmployeeID = 5  
RETURNING *;
```

Output

EmployeeID	Name
5	Arpit

The row is deleted and returned.

Very useful for:

- Audit logs
- Undo features
- Backups

Comparison: RETURNING vs SELECT

RETURNING	SELECT
Returns affected rows	Reads any rows
Works with INSERT/UPDATE/DELETE	Works independently
One query	Usually two queries

Internal Working

